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Primatology

Study of Primates

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Why Primates are Mammals?

- Primates belong to sub-class of mammals that show the presence of placenta (
- Presence of hair or fur on the body
- Presence of mammary glands.
- Presence of sweat and sebaceous glands
- Four types of teeth – Incisors, canines, premolars and molars

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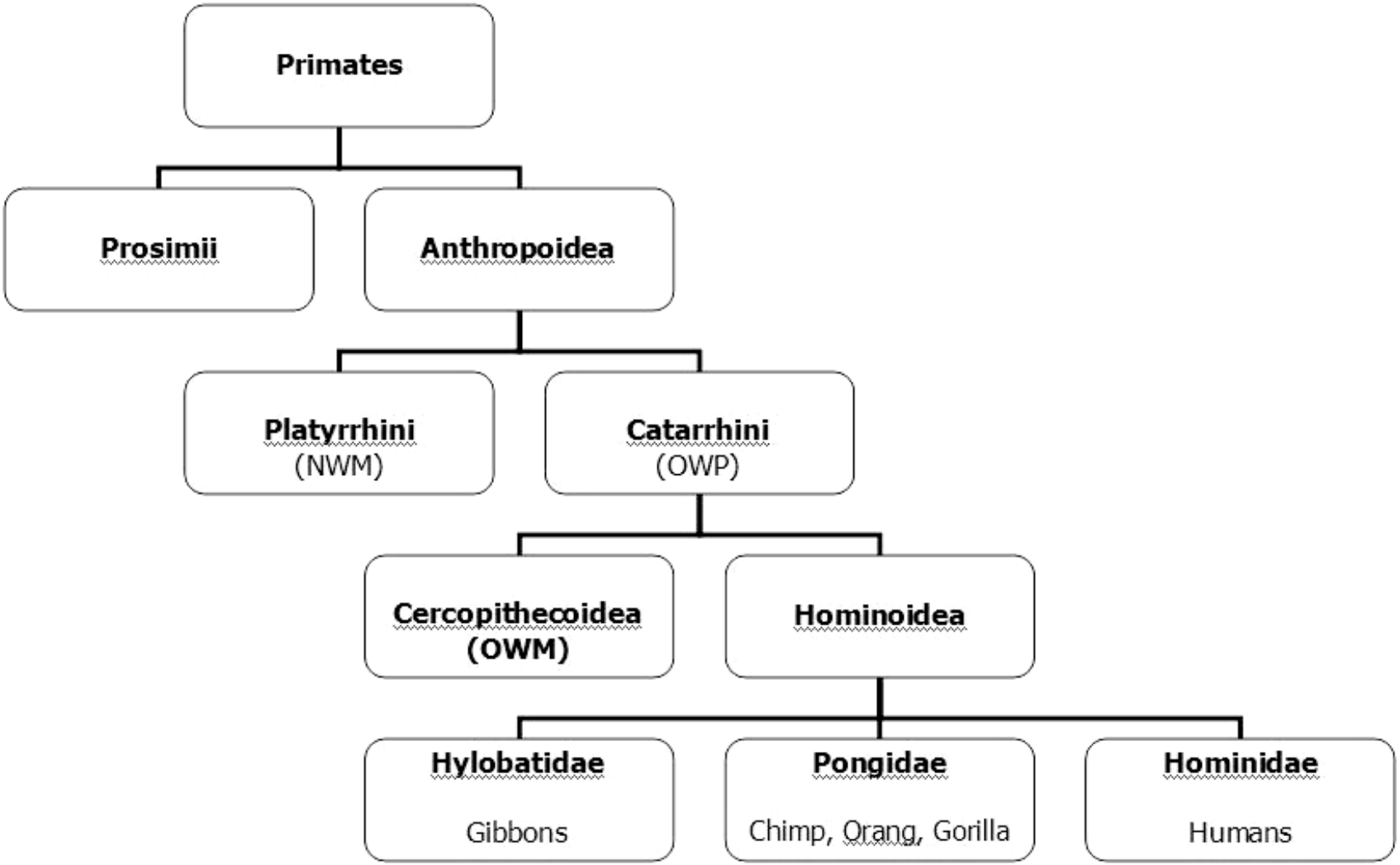
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Why Primates are Mammals?

- Presence of diaphragm which separates body cavity into anterior thorax and posterior abdomen
- Presence of external ear-pinna
- Aneucleated and circular RBCs
- Warm blooded nature – homeothermy
- Four chambered heart
- Vivipary, the development of young ones take place in uterus

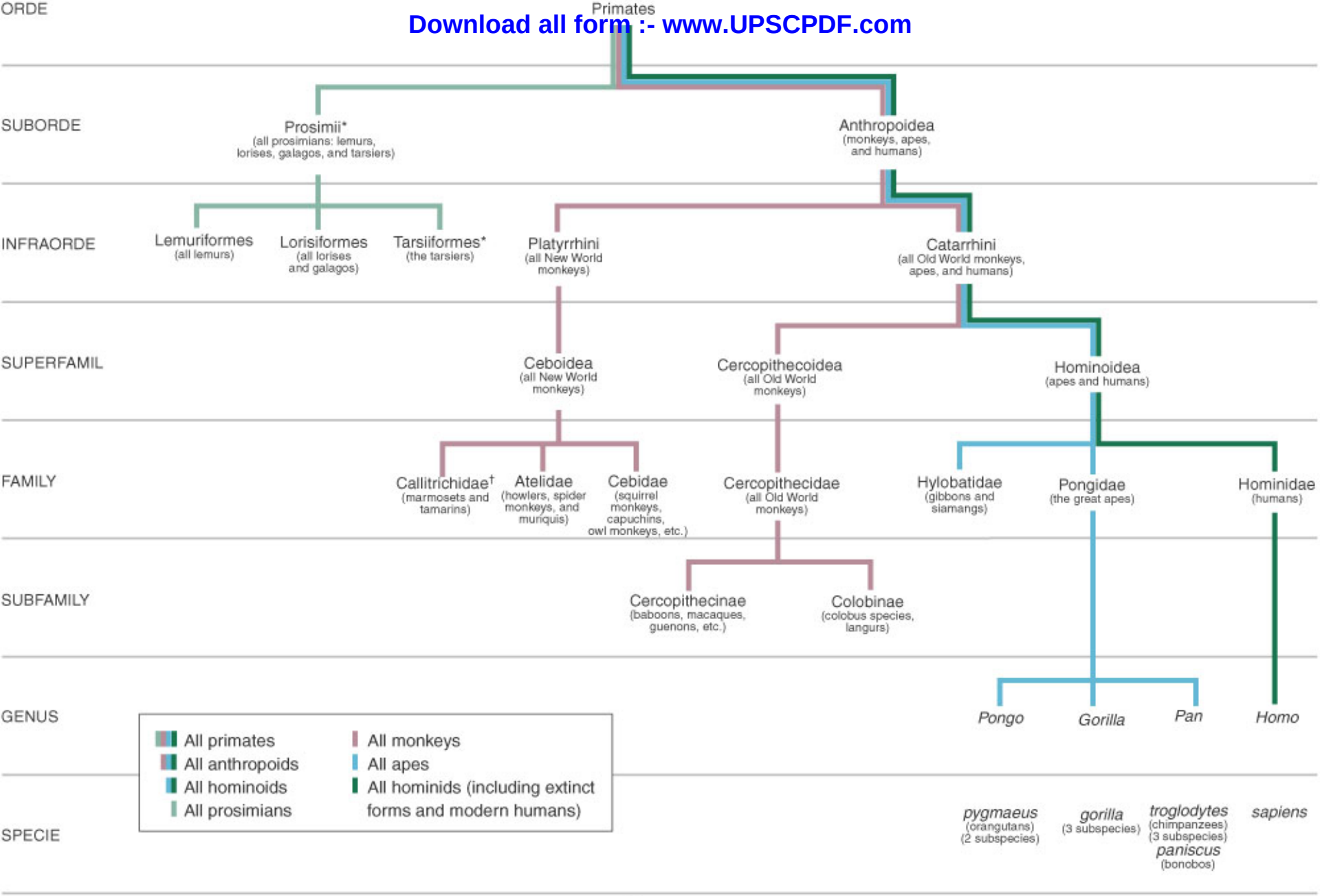
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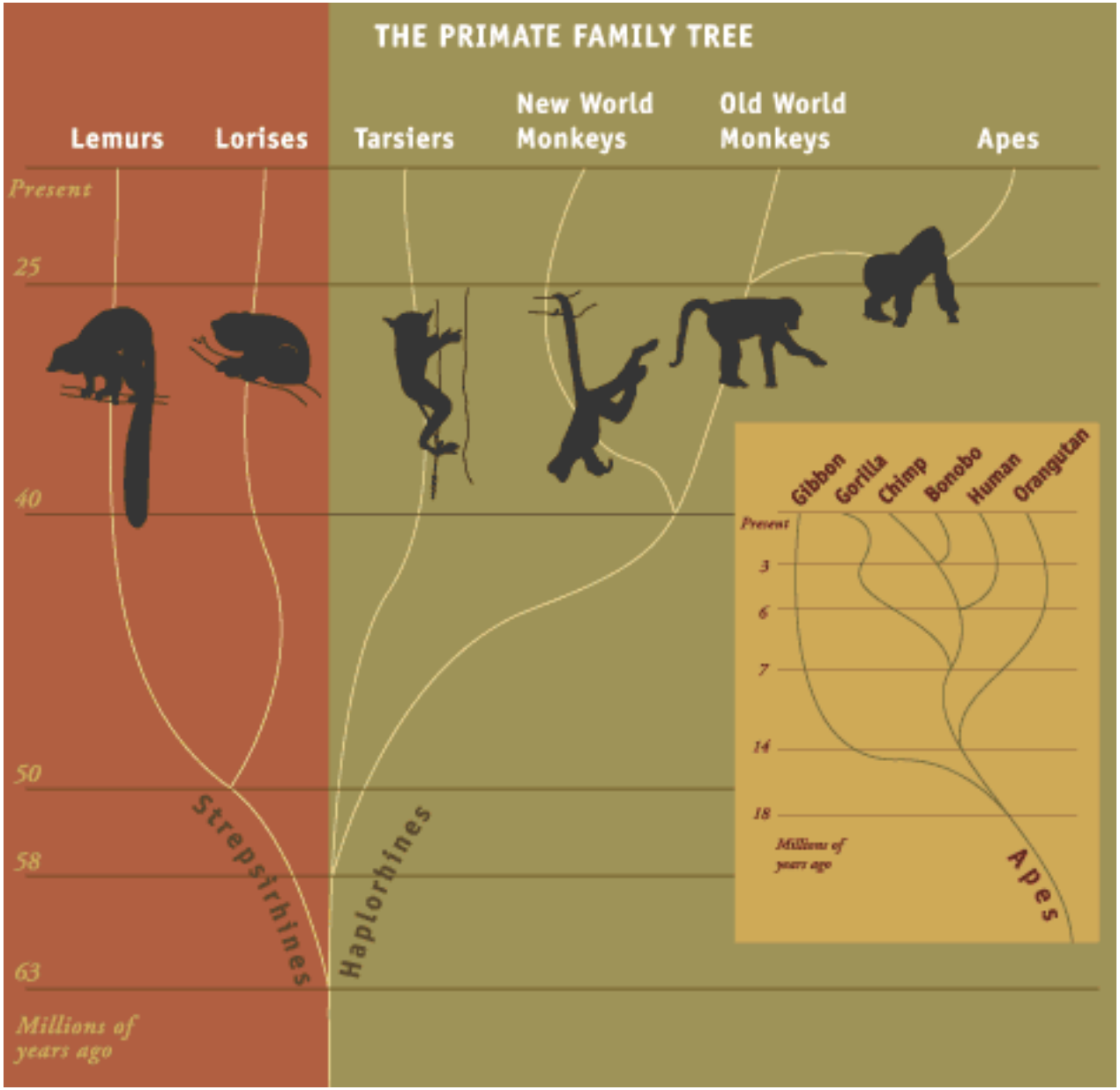
*There is some disagreement among primatologists concerning where to place tarsiers. Many researchers suggest that they more properly belong closer to the anthropoids and thus revise the primate classification to reflect this view. Here, for simplicity, we continue to use the traditional classifications.

†Fleagle (1999) and others have recently eliminated the family Callitrichidae and included marmosets and tamarins in the family Cebidae.

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Primate Evolutionary Tendencies

What makes a primate a primate?

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Why Study Evolutionary Trends?

- **What makes a primate a primate?**
- Defining primates like other taxonomical groups on the basis of morphological or anatomical characteristics is **extremely difficult - Why?**
  - Primates lack distinguishing bodily specializations like other mammalian groups
  - Primates exhibit amazing variety of forms and grades of organization - Size, Habitat, Behavior etc..
- The best and most useful solution to the problem of a definition was devised by one of the foremost primate anatomists, **Sir Wilfred E. LeGros Clark**.

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Nature of Trends

Though there are ten trends, there are related to three principle areas…

1. Limbs and extremities (1-3)
2. Head- eyes, brain and dentition (trends 4-8)
3. Life cycle-reproduction, growth, and longevity (trends 9 & 10)

- These are trends and not characteristics. Hence are not uniformly present, but are expressed to varying degrees among the members of the order.
- As a rule, except for the first one, these trends are less well developed in the living prosimians and in fossil forms than in the more advanced, modern Higher Primates.

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Nature of Trends

- They are all interrelated and adaptive to a mode of life which has been the outstanding factor in primate evolution-**the arboreal habitat**.
- Other mammals have taken up arboreal life but none have done it in exactly the same way.
- Many of man's characteristics are simply refinements of a basic primate pattern that evolved millions of years ago in response to the necessities of arboreal living.

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Evolutionary Trends

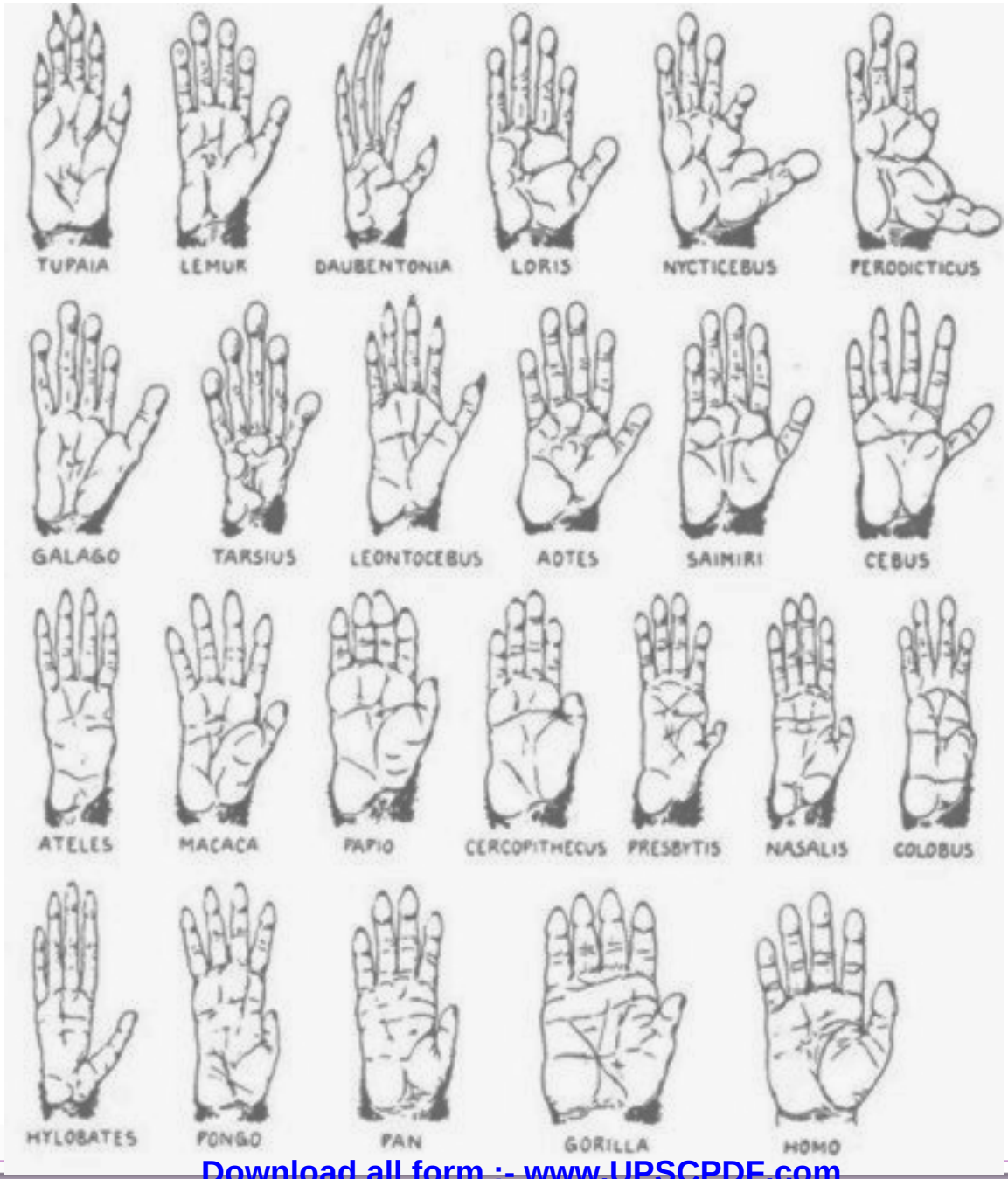
1. The preservation of a generalized structure of the limbs - like primitive pentadactyly, and the retention of certain elements of the limb skeleton (such as the clavicle) which tend to be reduced or to disappear in some groups of mammals.

2. An enhancement of the free mobility of the digits, especially the thumb and big toe (which are used for grasping purposes).

3. The replacement of sharp compressed claws by flattened nails, associated with the development of highly sensitive tactile pads on the digits.

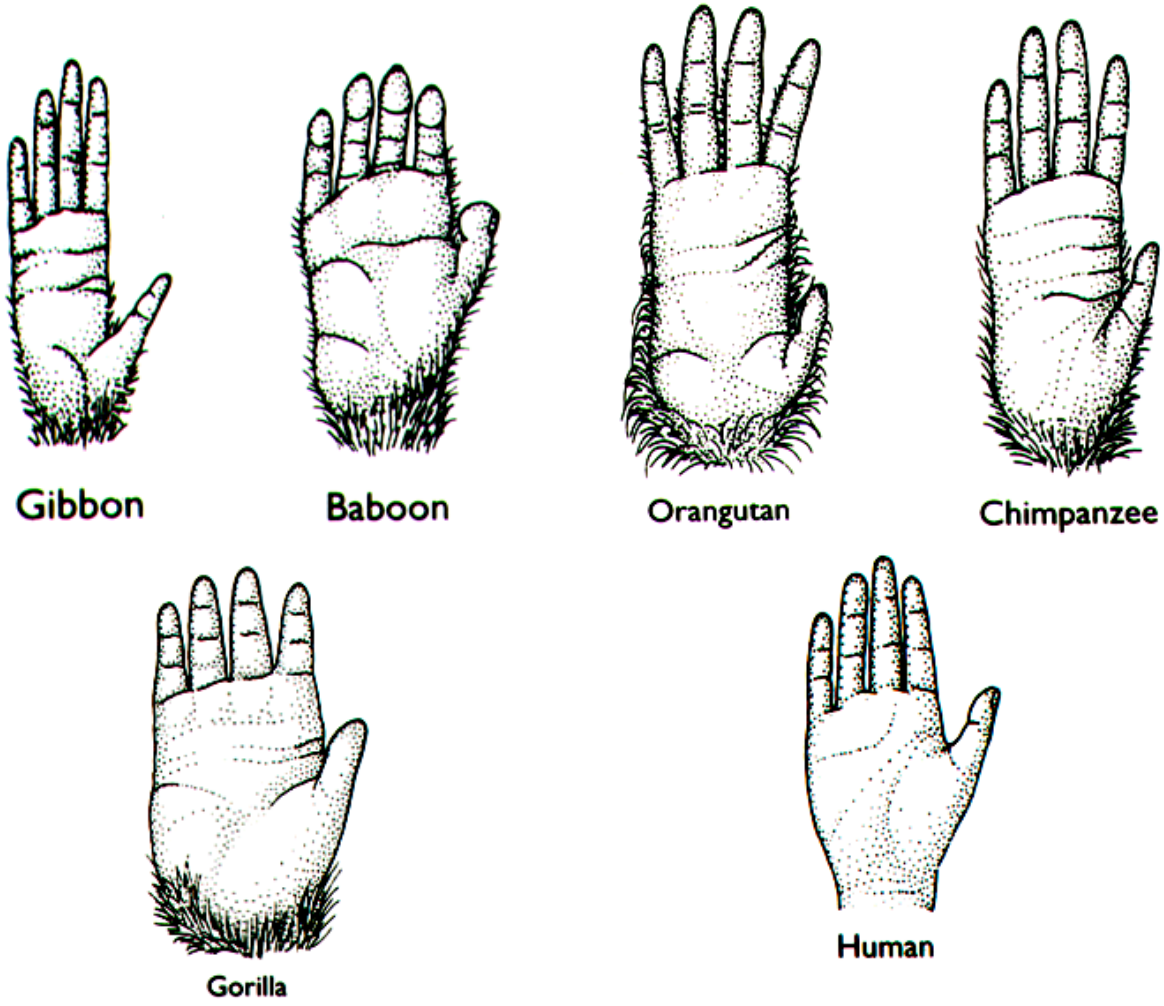
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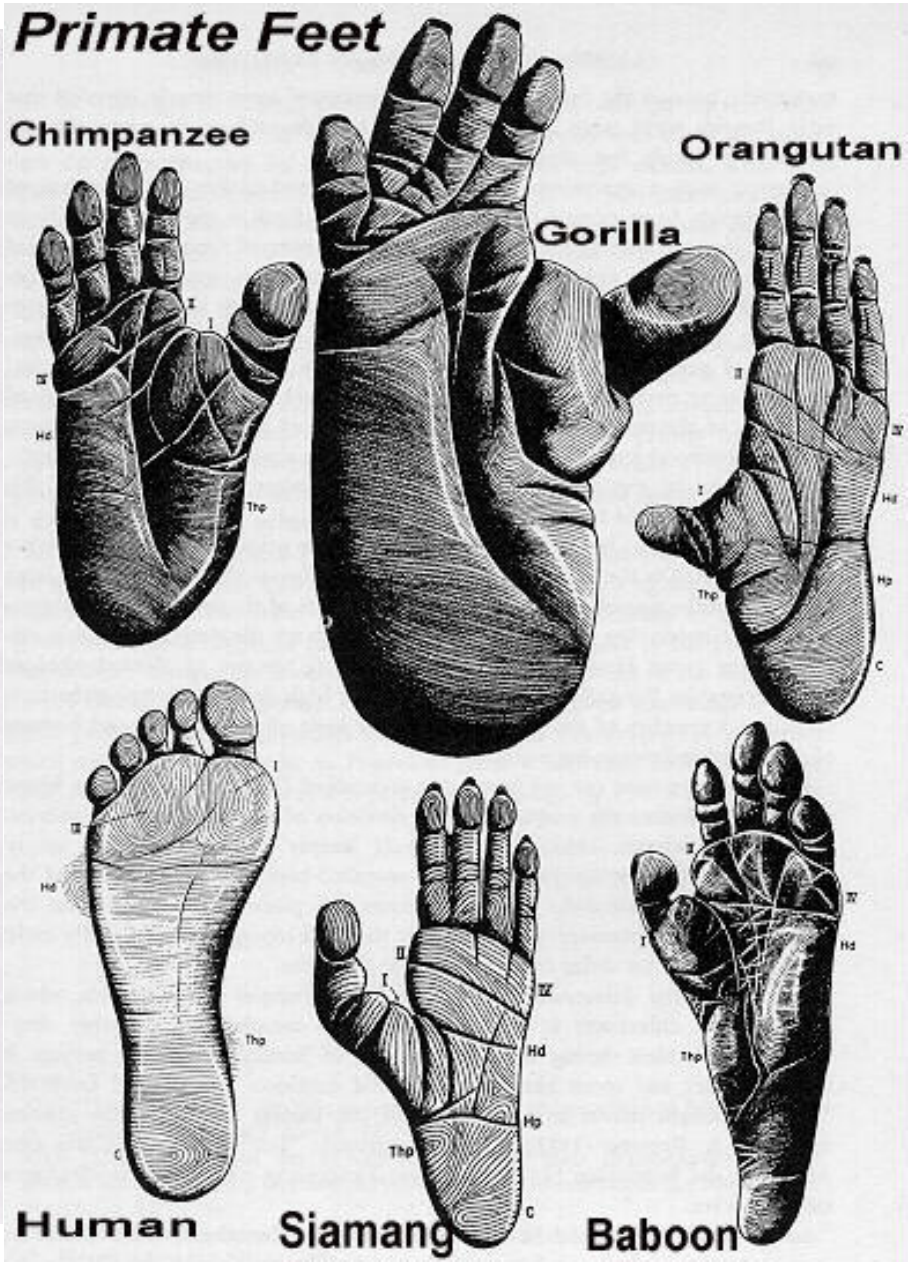
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Clavicles





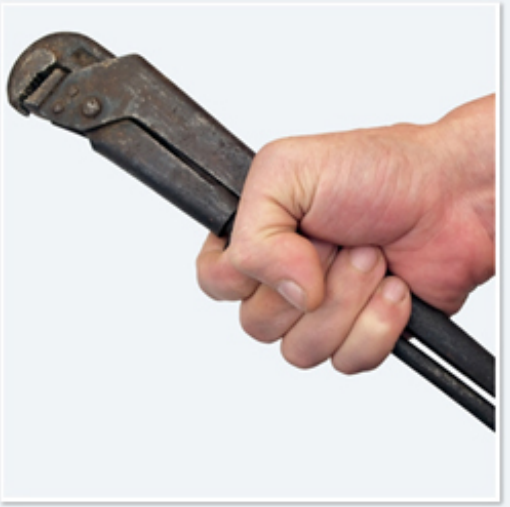
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Power & Precision Grip









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Thumb & Opposobility





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Nails & Friction Ridges

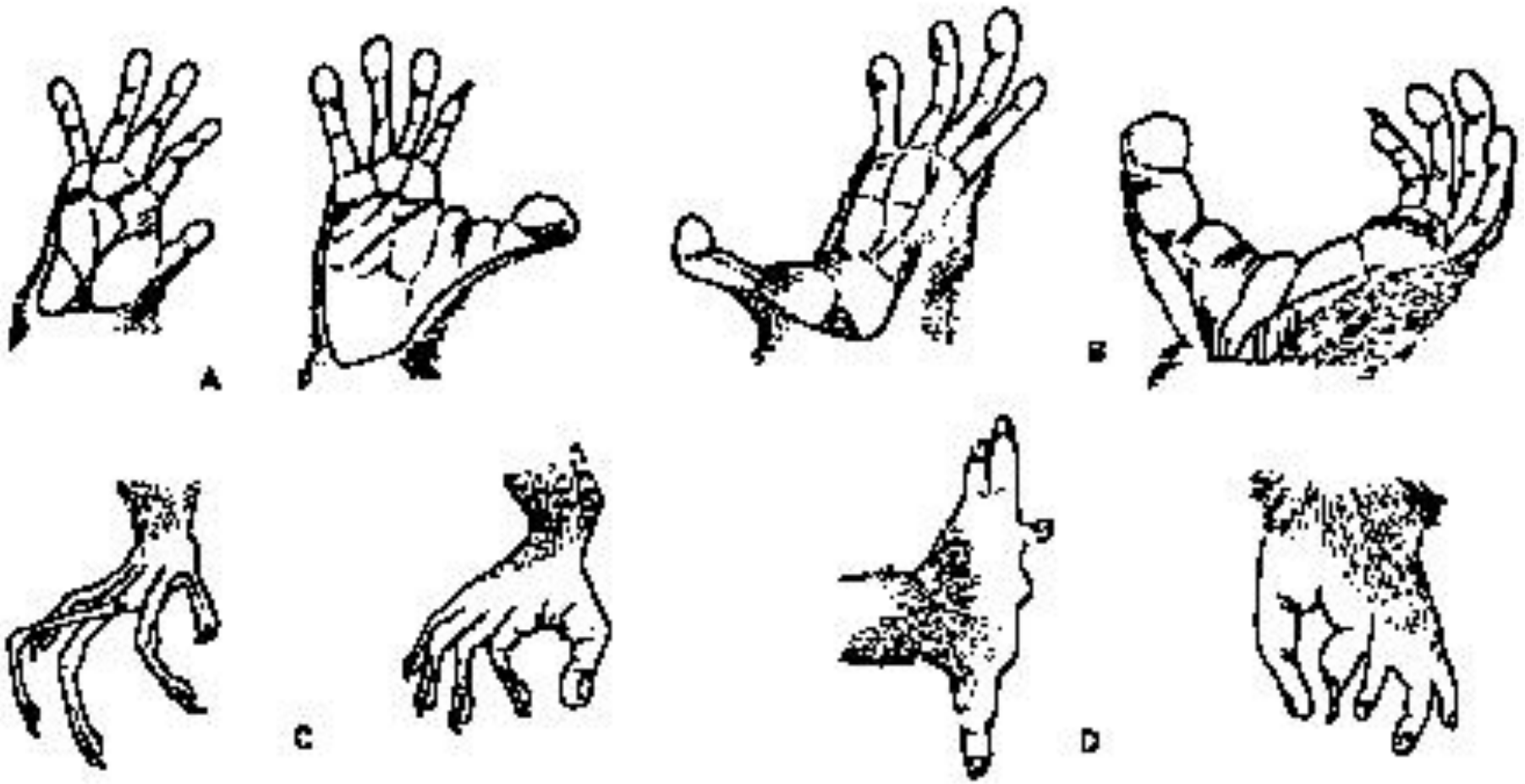




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Various levels of opposability



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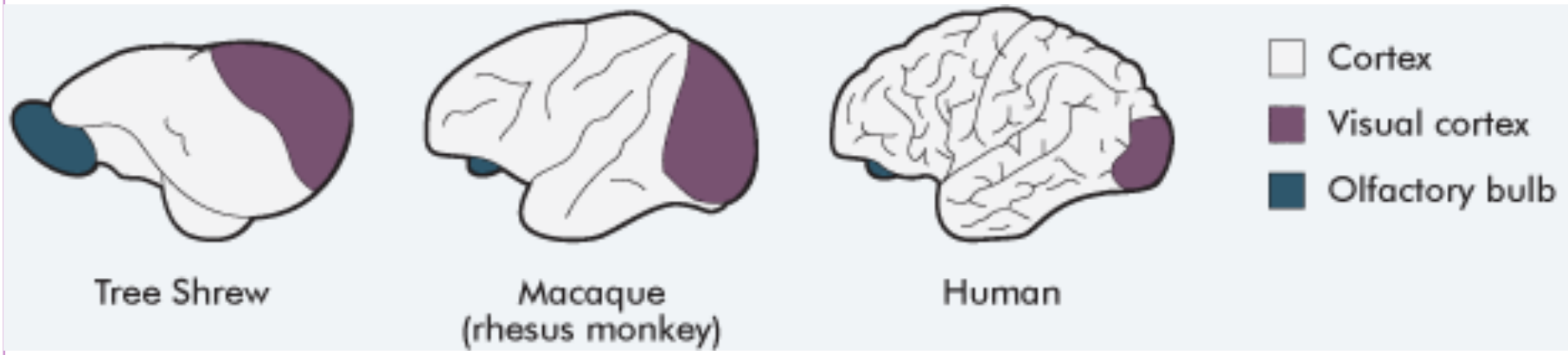
Primate Trends

4. The progressive abbreviation of the snout or muzzle.
5. The elaboration and perfection of the visual apparatus with the development to varying degrees of binocular vision.
6. Reduction of the apparatus of smell.

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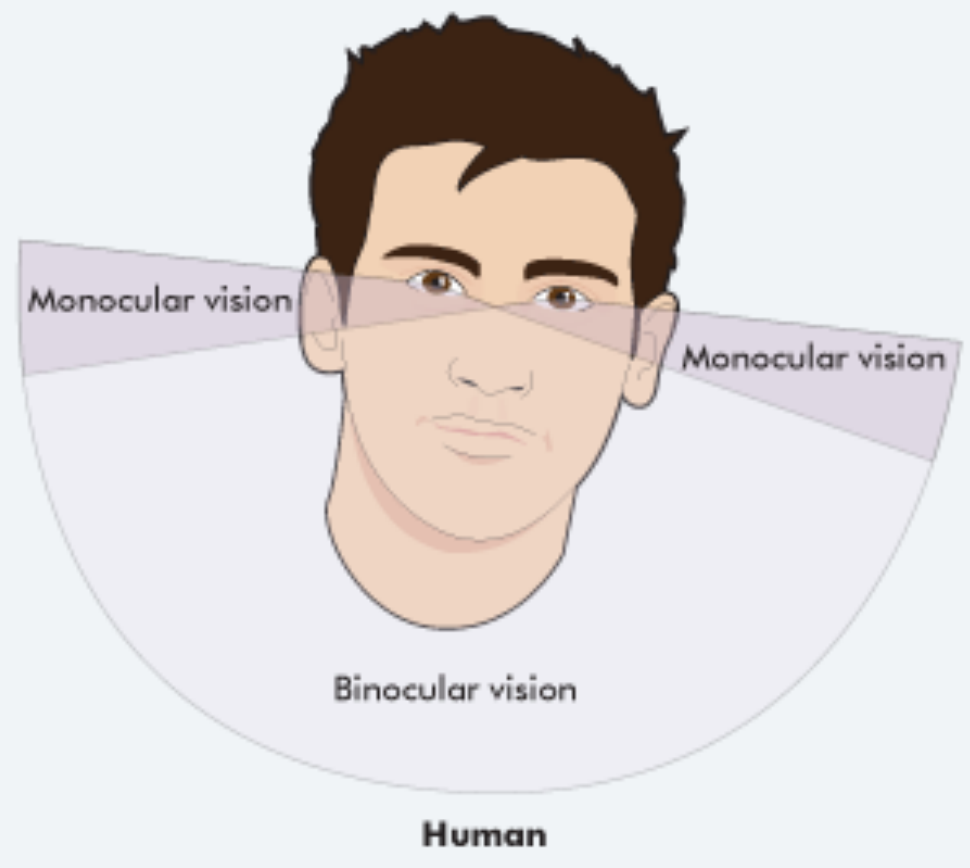
Reduction in Sense of Smell & Increase in visual cortex

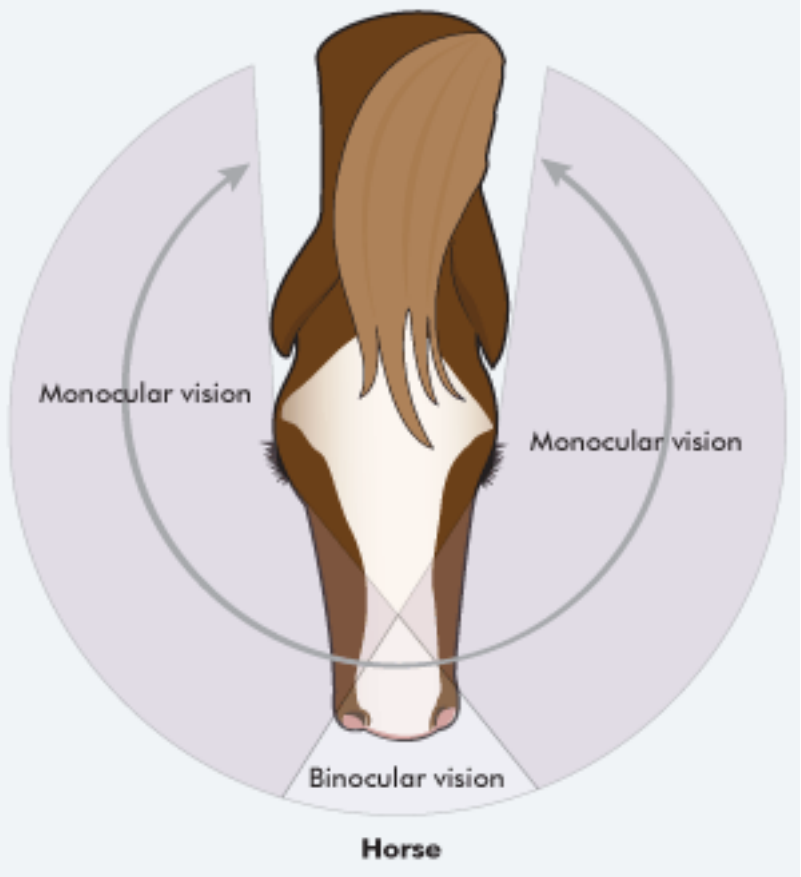


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Binocular Vision





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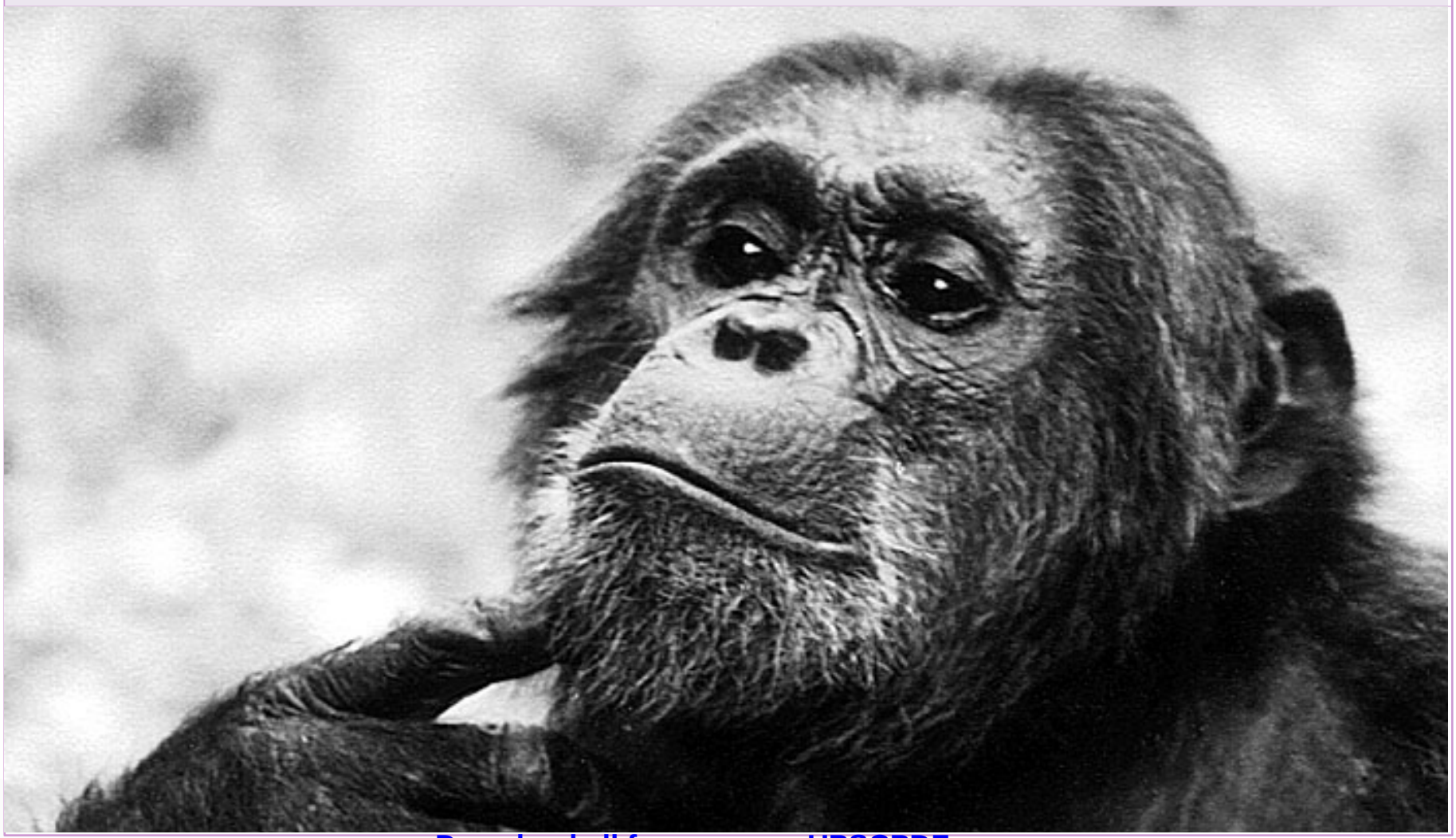
Stereoscopic Vision



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Front-facing orbits



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Primate Trends

7. The loss of certain elements of the primitive mammalian dentition (3:1:4:3), and the preservation of a simple cusp pattern of the molar teeth.

8. Progressive expansion and elaboration of the brain, affecting predominantly the cerebral cortex and its dependencies.

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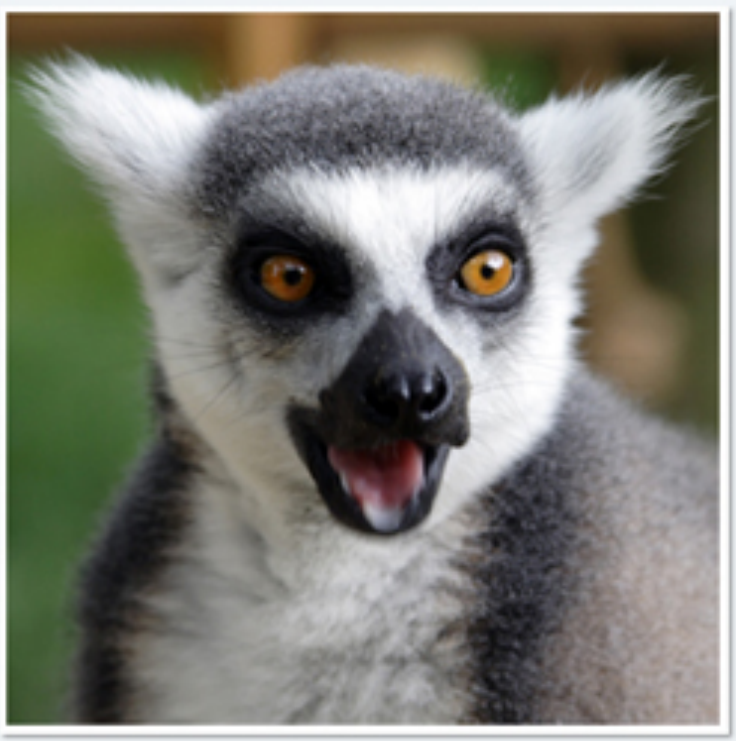
Dentition

Dental Formulae & Number of Teeth in Various Living Primates		
Group	**Dental Formula**	**No. of Permanent Teeth**
Lemuridae	2-1-3-3/2-1-3-3	36
Indridae	2-1-2-3/2-0-2-3	30
Lorisoidea	2-1-3-3/2-1-3-3	36
Tarsioidea	2-1-3-3/1-1-3-3	34
Cebidae	2-1-3-3/2-1-3-3	36
Callithricidae	2-1-3-2/2-1-3-2	32
Cercopithecoidea	2-1-2-3/2-1-2-3	32
Hominoidea	2-1-2-3/2-1-2-3	32

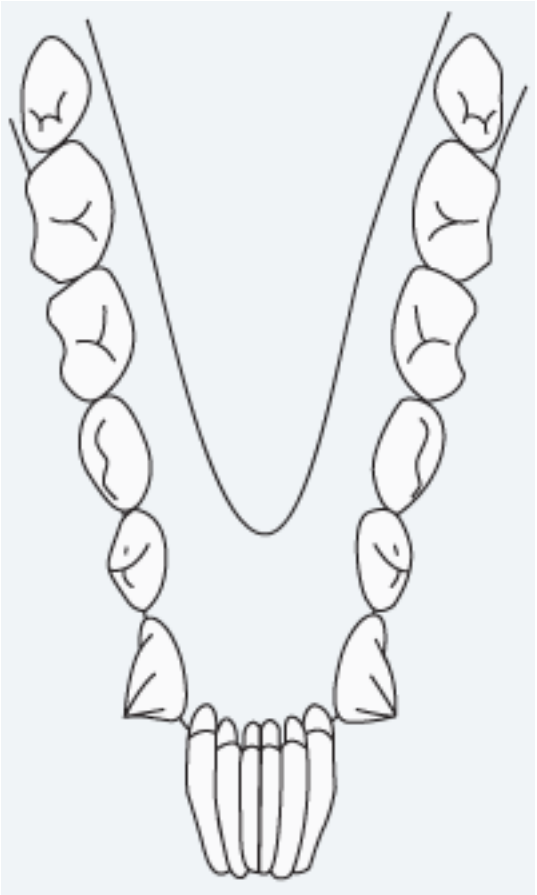
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Dental Comb in Lower Primates



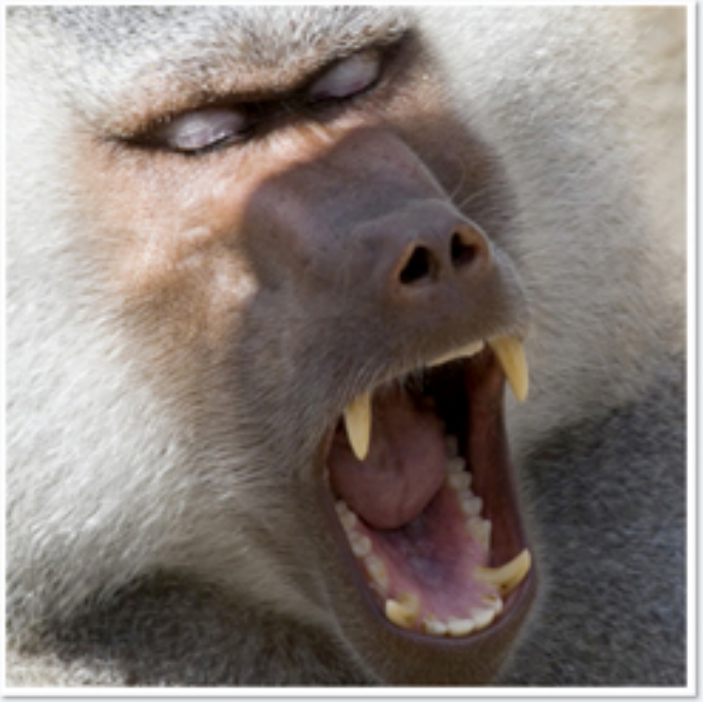
Lemur



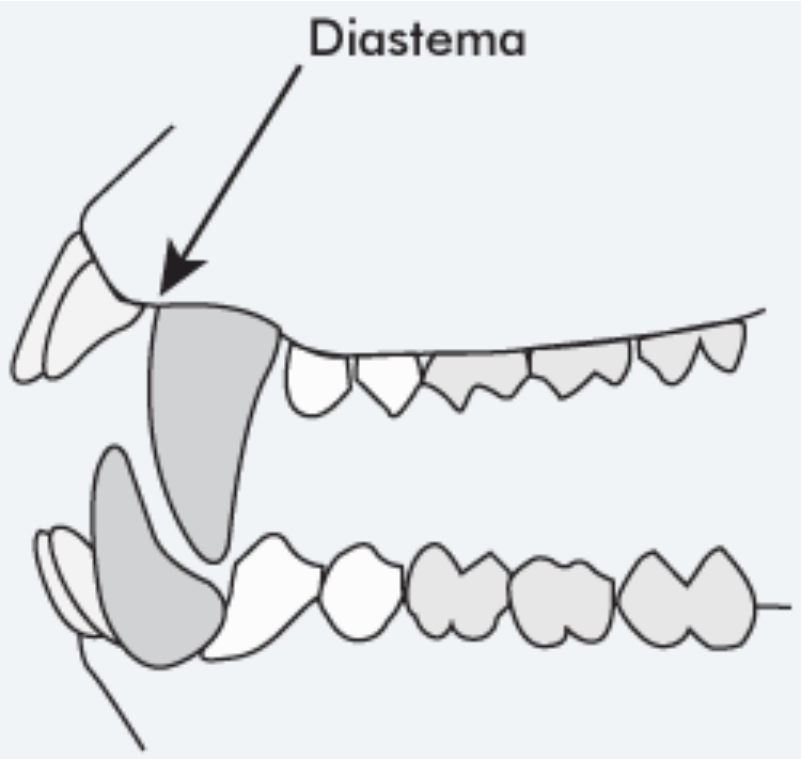
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Diastema



Baboon displaying teeth



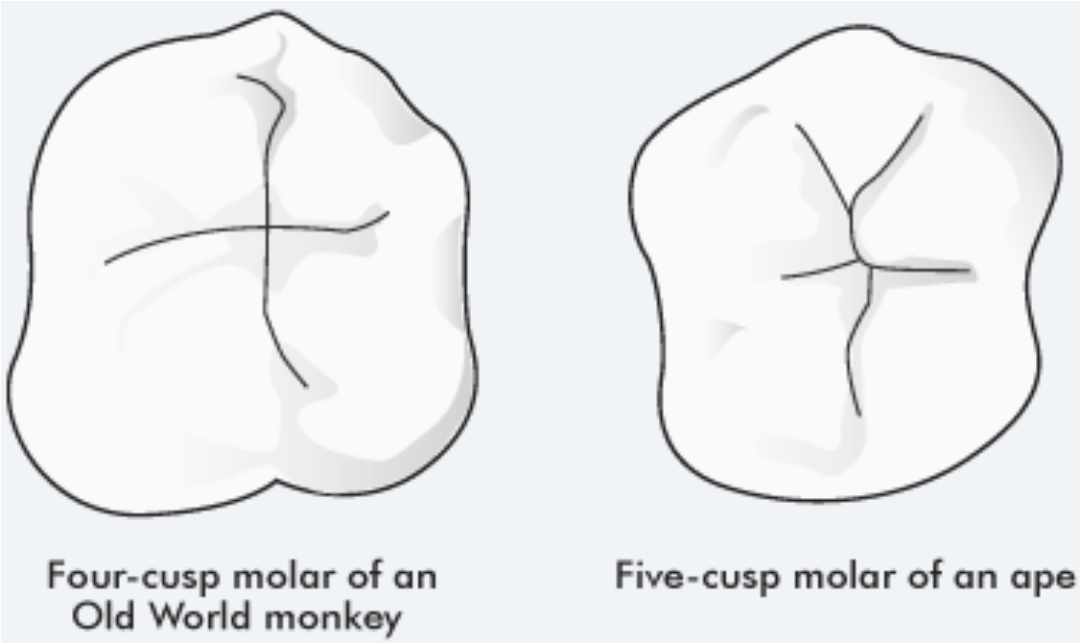
The diastema is a tooth gap found in Old World monkeys and apes.
A diastema is located between the upper second incisor and the upper canine to accommodate the large lower canine so the jaw can close comfortably.

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Molar Cusps

Old World monkeys have four cusps and apes and humans have five. It would seem that the increased number of cusps would be for a specialized diet of more plant material.



Raised surfaces of the molar teeth are cusps. When the upper and lower surfaces of these teeth come together they provide an excellent grinding surface for tough and fibrous, food-like plant material.

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Legacy haunts…

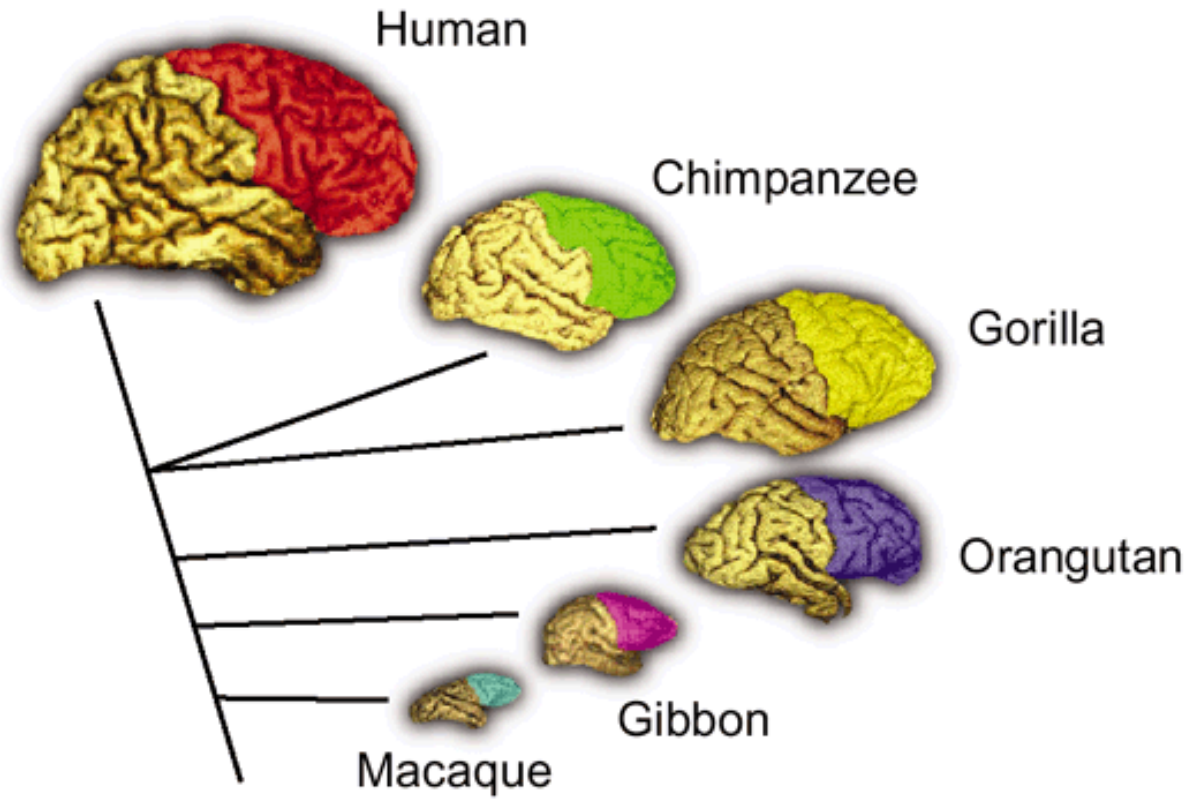




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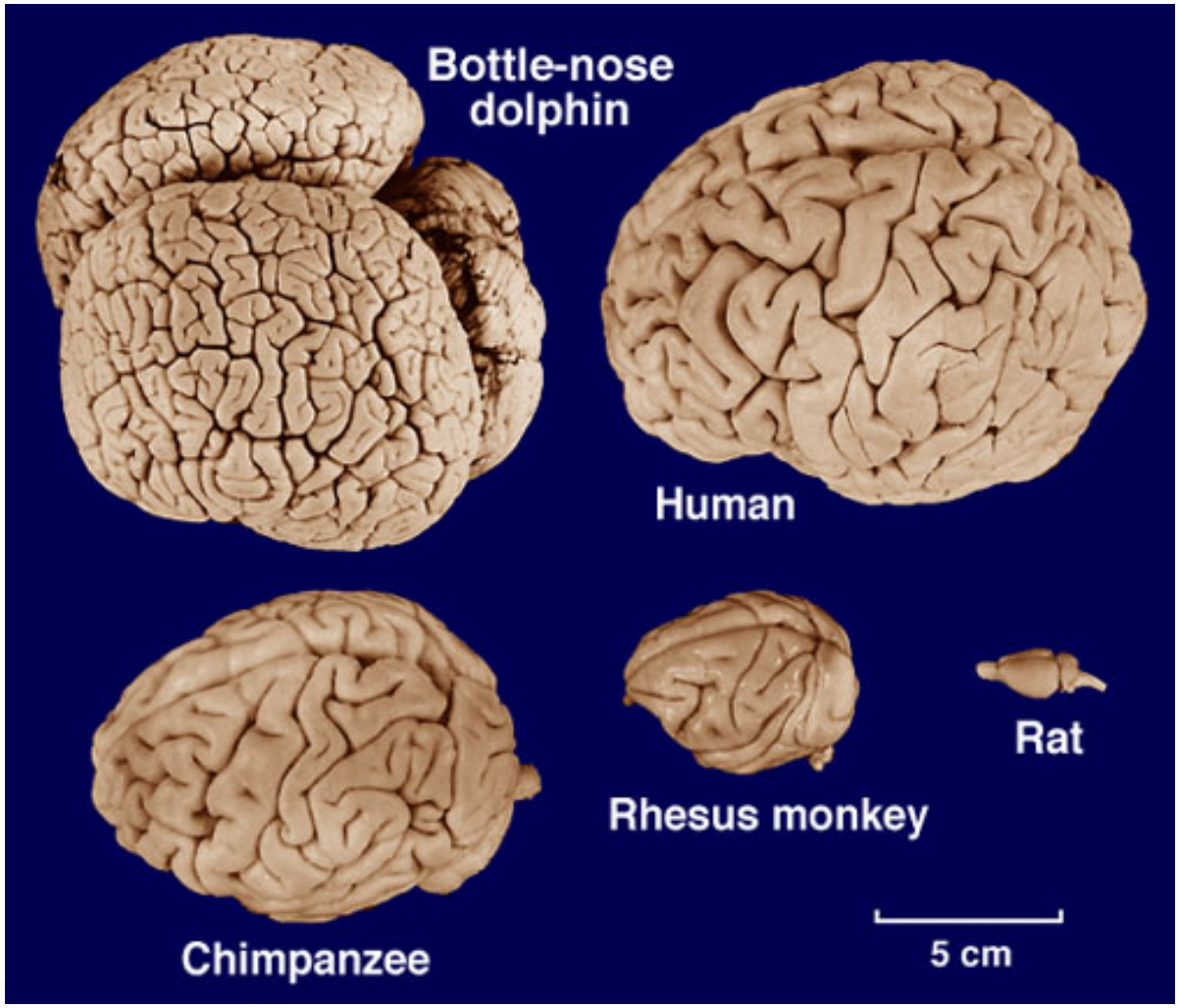
Increasing brain size & complexity in cortical area



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Brain



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Primate Trends

9. Progressive and increasingly efficient development of those gestational processes concerned with the nourishment of the fetus before birth.

10. Prolongation of postnatal life periods.

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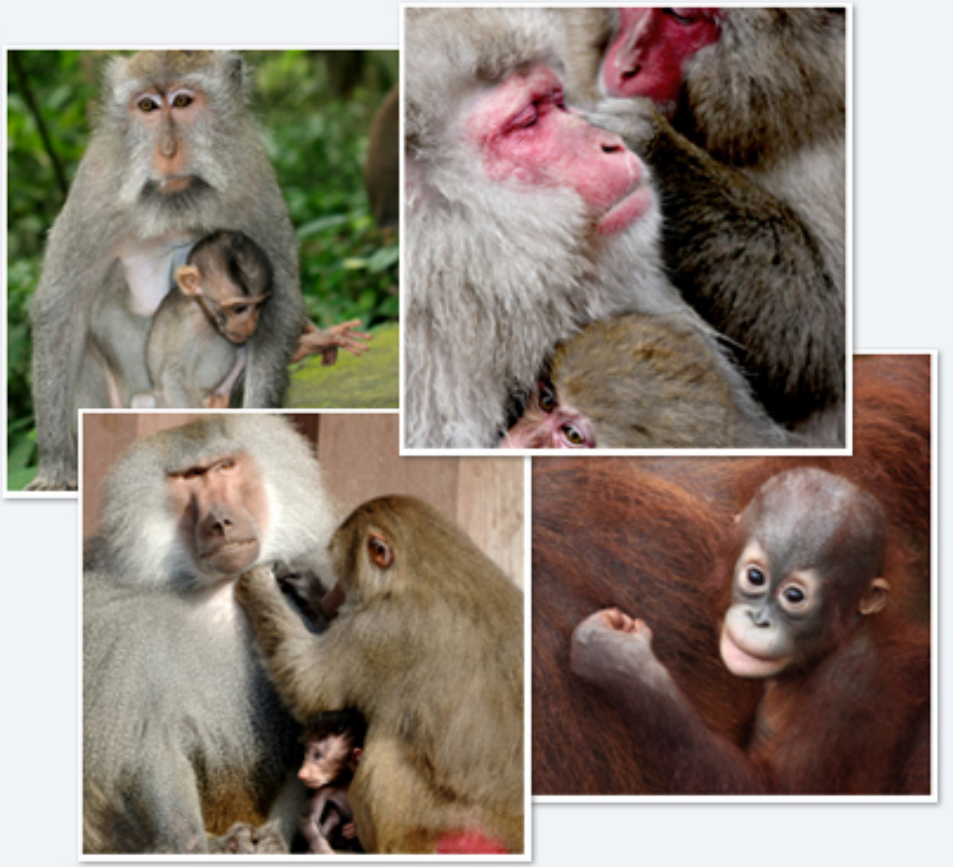
Growth & Development

Comparative Data on Growth & Development			
Species	**Gestation**	**Age at sexual maturity**	**Length of growth period**
Rhesus Monkey	5 ½ months	2-3 years	7-8 years
Chimpanzee	7 ½ months	8-9 years	11-12 years
Man	9 months	13-15 years	20-21 years

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Post Natal Dependence





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